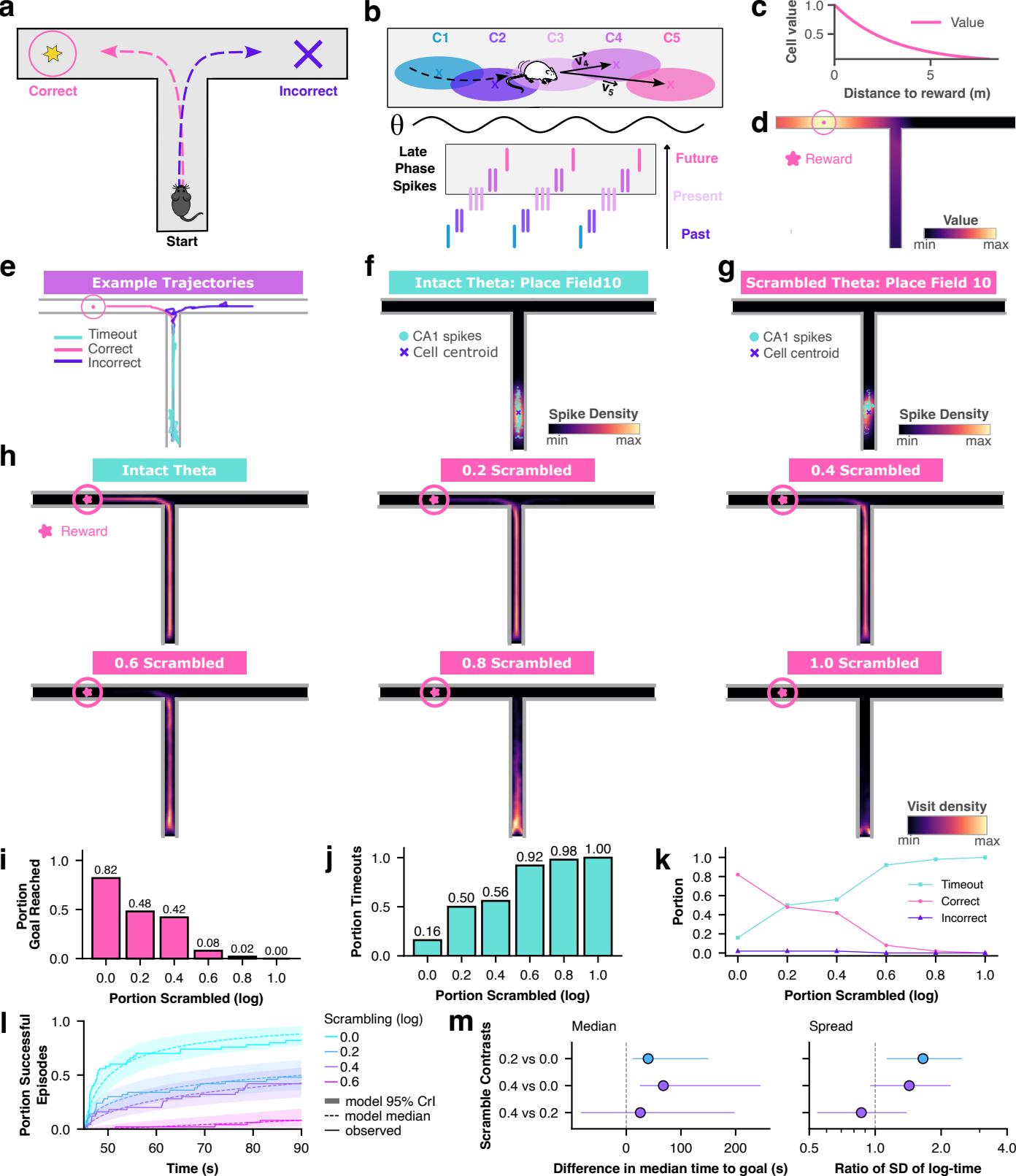

a
Correct
Incorrect
Start
b
C1
C2
C3
C4
C5
θ
Late Phase Spikes
Future
Present
Past
c
Cell value
1.0
0.5
0
5
Value
Distance to reward (m)
d
Reward
Value
min
max
e
Example Trajectories
Timeout
Correct
Incorrect
f
Intact Theta: Place Field10
CA1 spikes
Cell centroid
Spike Density
min
max
g
Scrambled Theta: Place Field 10
CA1 spikes
Cell centroid
Spike Density
min
max
h
Intact Theta
Reward
0.2 Scrambled
0.4 Scrambled
0.6 Scrambled
0.8 Scrambled
1.0 Scrambled
Visit density
min
max
i
Portion Goal Reached
0.82
0.48
0.42
0.08
0.02
0.00
Portion Scrambled (log)
j
Portion Timeouts
0.16
0.50
0.56
0.92
0.98
1.00
Portion Scrambled (log)
k
Portion
Timeout
Correct
Incorrect
Portion Scrambled (log)
l
Portion Successful Episodes
Time (s)
Scrambling (log)
0.0
0.2
0.4
0.6
model 95% CrI
model median
observed
m
Scramble Contrasts
0.2 vs 0.0
0.4 vs 0.0
0.4 vs 0.2
Median
Spread
Difference in median time to goal (s)
Ratio of SD of log-time